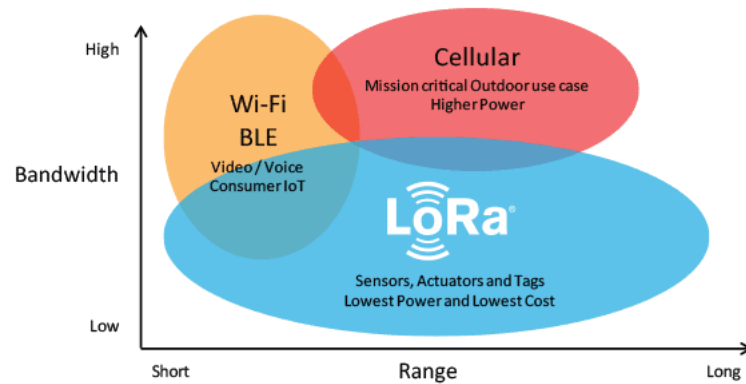


LoRa (Long Range, Low Power Wireless) and LoRaWAN (LoRa Wide Area Network)

Core Concept



LoRa = Long Range (Physical layer modulation technique)

LoRaWAN = Long Range Wide Area Network (Protocol/MAC layer)

LoRa Overview

A **long-range, low-power** wireless technology for:

- **Ultra-long distance** ($\approx$ 1.2–9.3 miles urban, 9.3–31 miles rural)
- **Very low power** (10+ years on battery)
- **Low data rates** (300 bps - 50 kbps)
- **Unlicensed ISM bands** (433/868/915 MHz)

Real analogy: Like a postal service for tiny messages - slow delivery but can reach remote villages that regular mail (WiFi/cellular) can't reach, and uses very little fuel (battery power)!

Feature	LoRa	WiFi	Cellular (LTE-M)	Bluetooth	Zigbee
Range	2-15 km	50m	1-10 km	10-100m	10-100m
Data Rate	0.3-50 kbps	50+ Mbps	1 Mbps	1-3 Mbps	250 kbps
Battery Life	10+ years	Hours-days	Days-weeks	Days-months	Months-years
Infrastructure	Optional gateway	Router required	Cell towers	No	Coordinator
Cost	Low (\$5-15)	Low (\$2-5)	Medium + fees	Low (\$2-5)	Low (\$5-10)
License	Free (ISM)	Free	Licensed	Free	Free
Security	AES-128	WPA2/3	Strong	Pairing	AES-128
Bandwidth	125-500 kHz	20-160 MHz	1.4-20 MHz	1-2 MHz	2 MHz
Best For	Rural sensors	Internet	Mobile tracking	Peripherals	Home automation

Use LoRa When:

- Need VERY long range (> 1 km)
- Battery must last YEARS (not rechargeable)
- Data is SMALL and INFREQUENT (temp every hour)
- No infrastructure available (rural, remote)
- Low cost per node critical
- Coverage over large area (farm, city)

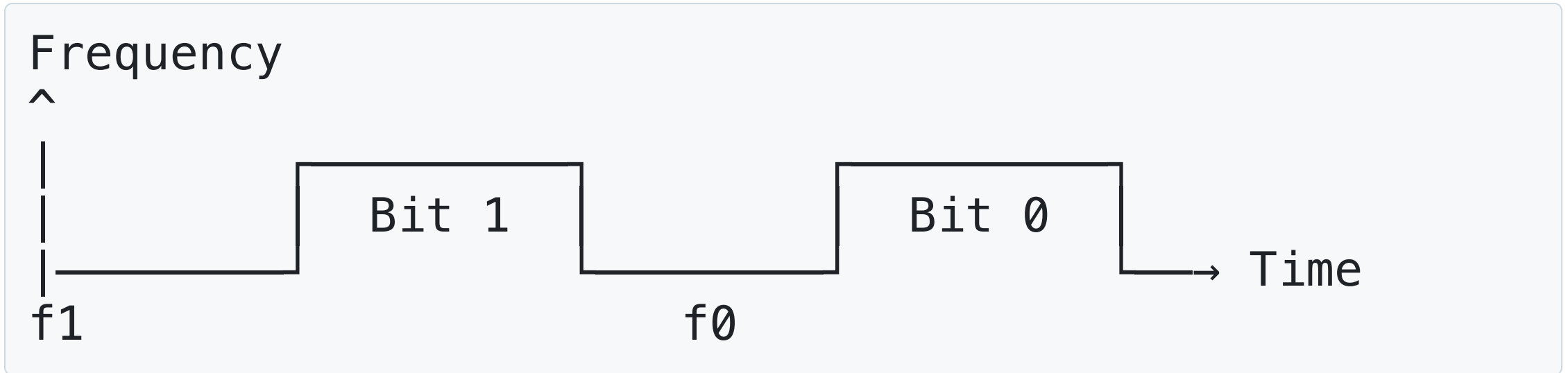
Examples:

- Agricultural soil sensors (100s of acres)
- Smart city parking sensors
- Remote tank level monitoring
- Asset tracking (shipping containers)
- Wildlife tracking collars
- Weather stations in mountains

How LoRa Modulation Works

Chirp Spread Spectrum (CSS)

Traditional Radio (FSK)



Single fixed frequency per bit

- Bit 1 → one constant frequency (f_1)
- Bit 0 → another constant frequency (f_0)
- Frequency stays stable during each bit period

It's like a shouting: *We can easily lost signal if noise or interference hits that frequency at the wrong time!*

LoRa (Chirp Spread Spectrum, CSS)



Frequency continuously sweeps during each **symbol**.

It's like a whistle: *Even if noise hits at the wrong time, the receiver can still detect the chirp pattern and recover the data!*

"Chirp" = frequency increases (or decreases) over time.

- Each symbol is a frequency sweep
- Information encoded in the timing/phase of the chirp
- Much more robust to noise and long-distance attenuation

Why it's called "chirp": Sounds like a bird chirping when you hear it!

LoRa Parameters

Spreading Factor (SF)

Spreading Factor (SF7 to SF12):

Higher SF = Longer range, slower speed, more power

SF7: Fast (5.5 kbps), Short range, Low power

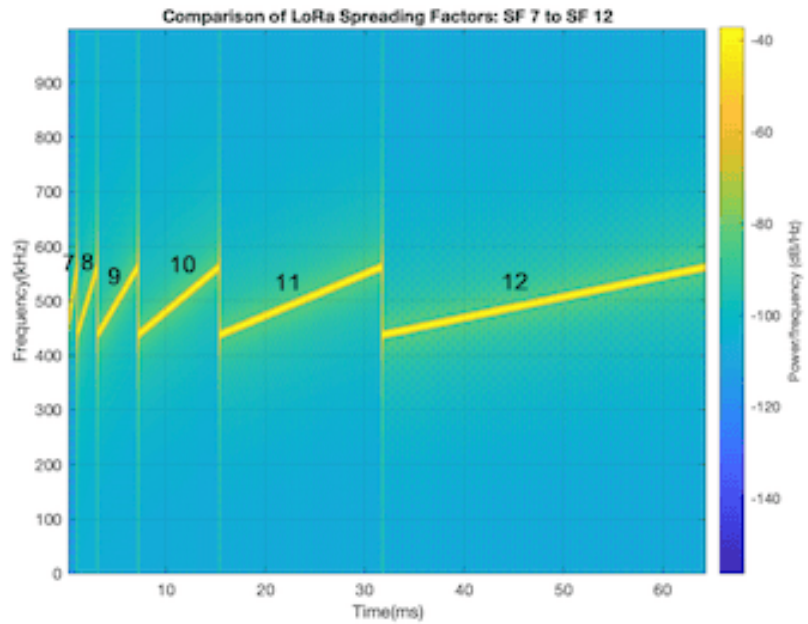
SF8: ↓

SF9: ↓

SF10: ↓

SF11: ↓

SF12: Slow (250 bps), Long range, High power



2^{SF} = Number of chirps per symbol

- More chirps = Longer symbol time = Better sensitivity = Longer range

Example: More SF to increase range, but decreases data rate:

SF7: 3 km range, 5.5 kbps
SF12: 15 km range, 250 bps

Each step up in SF:

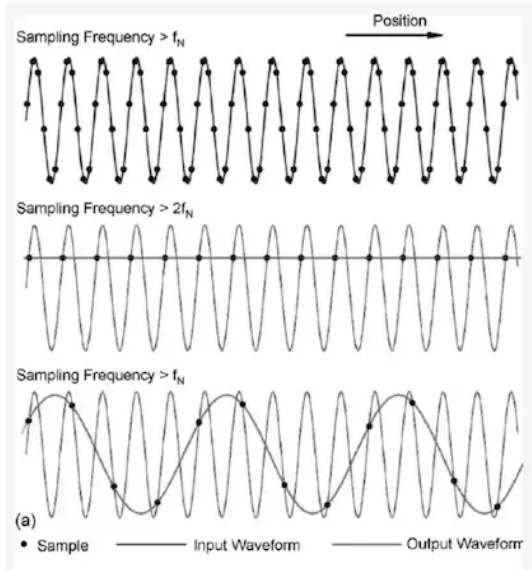
- Doubles transmission time
- Doubles power consumption
- Increases range by ~40%

SF (Spreading Factor) coupled with BW (Bandwidth) determines symbol time (T_{symbol}):

$$T_{symbol} = \frac{2^{SF}}{BW}$$

1. 2^{SF} = Number of frequency resolution count per one symbol.
2. BW = frequency range to sweep
3. Samples at a rate equal to the bandwidth
4. Wider bandwidth **requires** faster sampling (to catch high frequency signal we need to sample faster).

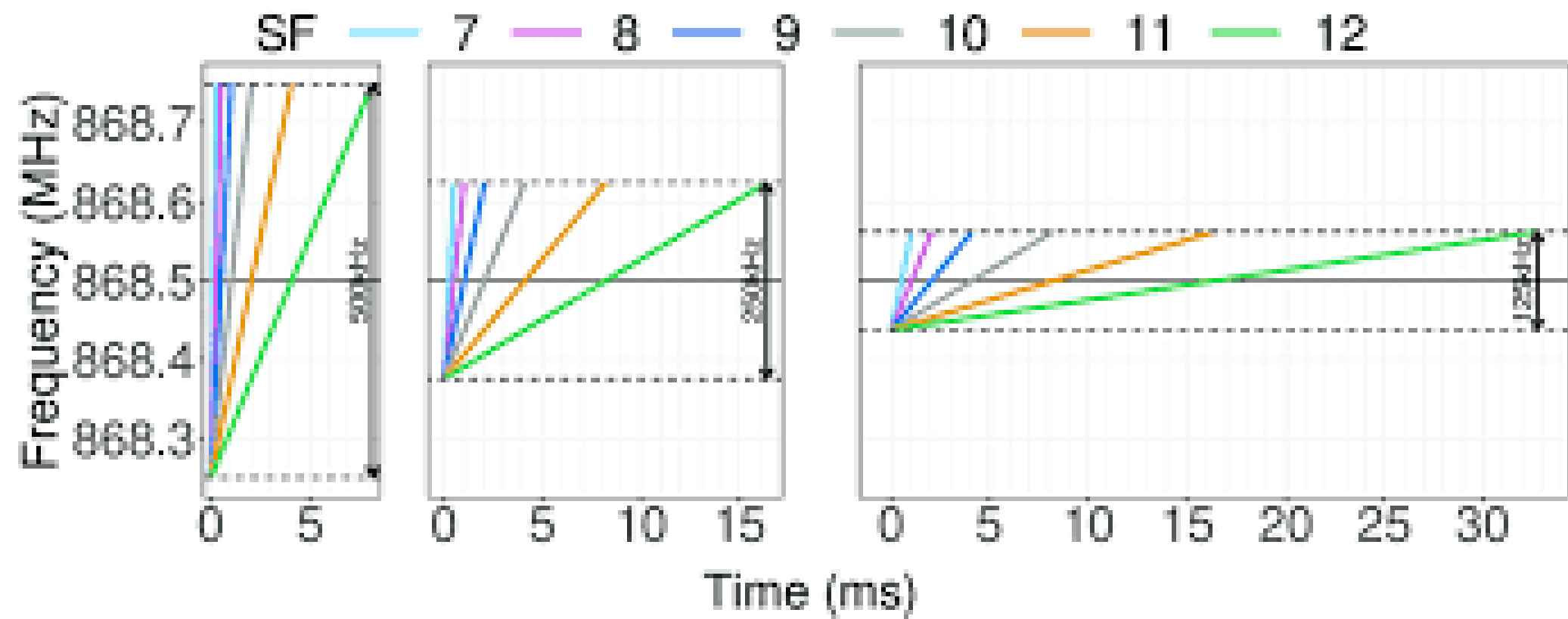
Understand frequency and sampling rate:



- For high-frequency signal (wider BW), we must sample faster to capture the signal accurately (Nyquist theorem).
- For low-frequency signal (narrower BW), we can sample slower.

Think about a whistle from freq A to B (BW), and I need to stop to sample the sound at certain points N times during the whistle:

- You need to sample faster for high frequency whistle (wider BW).
- To blow a whistle and sample wider BW, I need to blow faster and sample faster to catch the high frequency.
- To blow a whistle and sample narrower BW, I need to blow slower and sample slower because it's OK to sample slower for low frequency.
- Higher BW means faster data.
- However, wider BW means more noise and shorter range.



Bandwidth (BW)

Bandwidth (125 kHz, 250 kHz, 500 kHz):

Wider bandwidth = Faster data, shorter range

125 kHz: Longest range, slowest (default)

250 kHz: Medium range, medium speed

500 kHz: Shortest range, fastest

Trade-off:

Narrow BW: Better sensitivity, longer range

Wide BW: Faster transmission, more interference

Coding Rate (CR)

To recover from errors, LoRa adds redundancy using Forward Error Correction (FEC).

4/5: Less overhead (20%), faster, less robust

4/6: ↓

4/7: ↓

4/8: More overhead (100%), slower, more robust

Higher CR = More error correction = Better reliability in noisy environments

LoRa Packet Structure

Preamble 8 bytes	Header 0–20 B	Payload 0–255 B	CRC 2 bytes	
---------------------	------------------	--------------------	----------------	--

Preamble: Synchronization pattern

Header: Contains payload length, coding rate, CRC

Payload: Your actual data (sensor readings, commands)

CRC: Error detection

Example packet (temperature sensor):

Preamble	Header	Temp:23.5C,Humidity:65%	CRC
----------	--------	-------------------------	-----

Total: $8+4+24+2 = 38$ bytes

- At SF7, BW125: ~70 ms transmission time
- At SF12, BW125: ~2.5 seconds transmission time!

Important: Maximum payload is 255 bytes, but smaller is better for:

- Faster transmission
- Less power consumption
- Lower collision probability

LoRa vs LoRaWAN Security

LoRa (Point-to-Point)

Basic LoRa modules (SX1276, RFM95):

- NO built-in encryption
- Data transmitted in clear text
- Anyone with receiver can decode
- Must implement your own encryption!

How to secure:

- Add AES encryption in firmware
- Use unique keys per device
- Implement challenge-response auth
- Add message counters (prevent replay)

LoRaWAN (Network Protocol)

LoRaWAN network:

- AES-128 encryption (built-in)
- Two-layer security: - Network Session Key (NwkSKey) - Application Session Key (AppSKey)
- Device authentication
- Frame counter (prevents replay attacks)
- Message Integrity Check (MIC)

More secure but requires:

- LoRaWAN gateway (\$300-1000)
- Network server
- Device provisioning

LoRa vs LoRaWAN Architecture

LoRa Point-to-Point (Simple)

Node 1 $\longleftrightarrow$ Node 2
(Arduino + LoRa) (Arduino + LoRa)

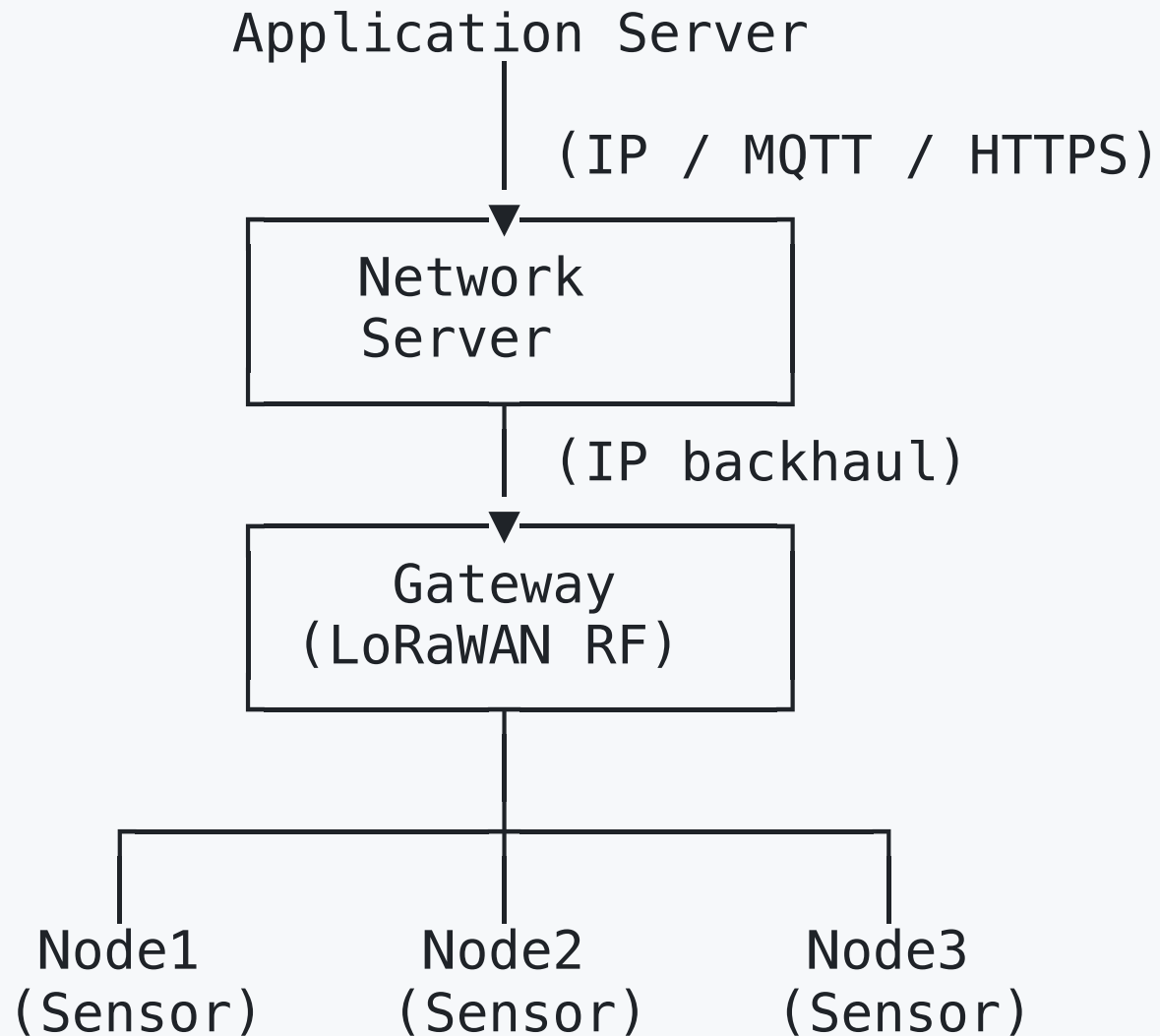
Pros:

- Simple to implement
- Low cost (no gateway)
- Direct communication
- Full control

Cons:

- Must implement protocol
- No built-in security
- Limited scalability
- No internet connectivity

LoRaWAN Network (Complex)



Pros:

- Built-in security (AES-128)
- Scalable (1000s of nodes)
- Internet connectivity
- Standard protocol

Cons:

- Needs gateway (\$300-1000)
- More complex setup
- Potential network fees (TTN is free)

LoRa Frequency Bands (Regional)

ISM Bands by Region

North America (USA, Canada):

- 902-928 MHz (915 MHz center)
- Allowed TX power: +30 dBm (1W)

Europe:

- 863-870 MHz (868 MHz center)
- Allowed TX power: +14 dBm (25 mW)
- Duty cycle limits: 1% (36 sec/hour)

LoRa Performance Analysis

Line of Sight (Flat terrain, no obstacles)

- **SF7:** 6–9 miles
- **SF12:** 19–31 miles
- **World Record:** ~517 miles (balloon, SF12)

Urban (Buildings, interference)

- SF7: 1.2–3 miles
- SF12: 3–6 miles

Indoor (Through walls)

- SF7: 0.06–0.19 miles (≈ 100 –300 m)
- SF12: 0.19–0.62 miles (≈ 300 –1000 m)

Factors affecting range:

- Spreading Factor (most important)
- Antenna height (2x height = 1.4x range)
- Antenna gain (3 dB = 2x power)
- TX power (20 dBm vs 14 dBm = 4x power)
- Obstacles (buildings, trees, hills)
- Frequency (lower = better propagation)
- Weather (rain absorbs signal)

Data Rate vs Range Trade-off

Configuration Data Rate Range Air Time (255 bytes)

- SF7 + BW125 + CR4/5 5.5 kbps Short 370 ms
- SF9 + BW125 + CR4/5 1.8 kbps Medium 1.1 sec
- SF12 + BW125 + CR4/5 250 bps Long 7.8 sec

Key insight: Range comes at the cost of speed!

Power Consumption Analysis

LoRa Module Current Draw

SX1276/RFM95W:

- Sleep mode: 0.2 μA (0.0002 mA)
- Standby: 1.5 μA (0.0015 mA)
- RX continuous: 10-12 mA
- TX +20 dBm (100 mW): 120 mA
- TX +14 dBm (25 mW): 30 mA
- TX +7 dBm (5 mW): 20 mA

Battery Life Calculation

Example: Agricultural sensor

- 3.7V 2500 mAh battery
- Send 20 bytes every 30 minutes
- SF12, BW125, CR4/5
- TX power: 20 dBm

Transmission time: ~180 ms

- TX current: 120 mA
- TX energy: $120 \text{ mA} \times 0.18 \text{ sec} = 21.6 \text{ mA}\cdot\text{sec}$

Sleep current: 0.01 mA (with voltage regulator)

- Sleep time: 1800 sec
- Sleep energy: $0.01 \text{ mA} \times 1800 \text{ sec} = 18 \text{ mA}\cdot\text{sec}$

Total per cycle: $21.6 + 18 = 39.6 \text{ mA}\cdot\text{sec}$

- Cycles per hour: 2
- Energy per hour: $79.2 \text{ mA}\cdot\text{sec} \approx 0.022 \text{ mAh}$

Battery life: $2500 \text{ mAh} \div 0.022 \text{ mAh/h} = 113,636 \text{ hours}$
 $= 4,735 \text{ days} = 13 \text{ YEARS!}$

Reality: $\sim 5\text{-}10$ years (accounting for battery degradation)

